Quantum walk–based stock option pricing (vs a dynamic programming approach)

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🚀 Motivation and Challenges

- Stock price X_k evolves as a **random walk**:
$$X_{k+1} = X_k + W_k, k = 0,1,2, \ldots$$
- W_k are i.i.d. random variables with finite mean.
- Independent of initial price X_0.
- Option: **buy 1 share at fixed price** K.
- **Time horizon:** N days.
- **Profit if exercised on day k at price x:**
$$\text{Profit} = x - K$$
- Can exercise **at most once**.
- **Goal:** Maximize expected profit.

🚀 Motivation and Challenges

- Let $V_m(x)$ =maximum expected profit with **m days left** and current price x.
- **Recursion:**

$$V_m(x) = \max(\, x - K,\ \mathbb{E}[V_{m-1}(x + W_{m-1})])$$

- **Structural insights:**
  - $V_m(x) - x$ is **non-increasing** in x.
  - $V_m(x)$ is **decreasing in m**.
- **Conclusion:** Optimal policy is **threshold-based:**

  Exercise *on day m if and only if* $x \geq p_m$
- Thresholds decrease over time:

$$p_0 \geq p_1 \geq \cdots \geq p_{N-1} \geq p_N$$

🚀 Quantum Walk Concept

- **Goal:** Estimate stock price transitions for option exercise using **quantum sampling**.
- **Stock model:** Random walk

$$X_{k+1} = X_k + W_k, W_k \sim Exp(2) - 0.5$$

- **Quantum walk idea:**
  - Discretize prices into **bins** (unary encoding, one qubit per bin).
  - Use **coin qubit** to decide movement:
    - $|0\rangle \rightarrow$ move left (price down)
    - $|1\rangle \rightarrow$ move right (price up)
- **Output:** Quantum-sampled **transition matrix**

🚀 Quantum Walk Circuit

- **Registers:**
  - Coin qubit: controls left/right moves
  - Position qubits: unary encoding of bins
- **Coin operator C:**

```
qc.ry(theta, 0)  # theta = pi/2 for symmetric walk
```

- **Shift operator S:** controlled swaps of position qubits:

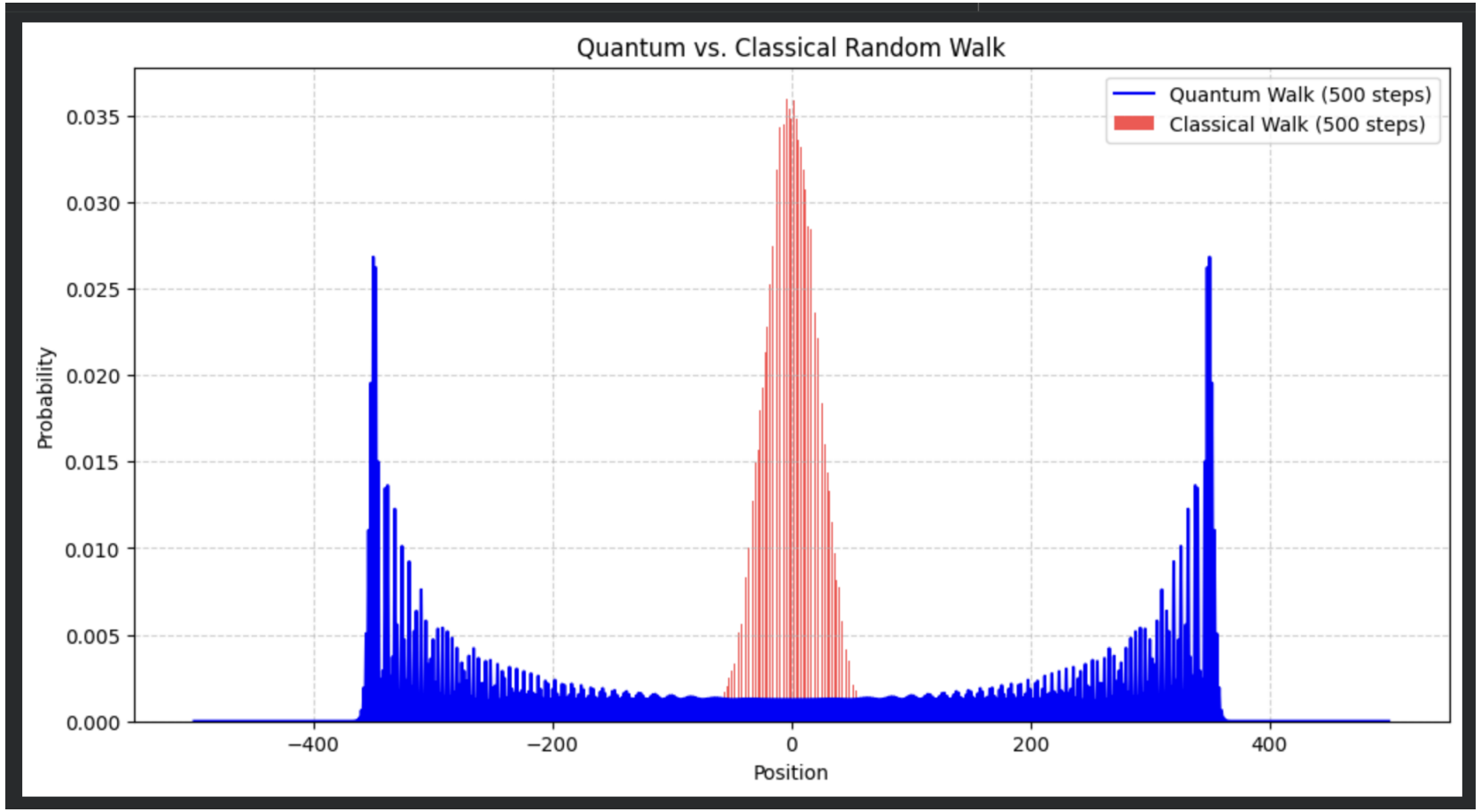
```
# Right move (coin == |1>)
for i in reversed(range(n_bins - 1)):
    cswap_between(qc, 0, 1 + i, 1 + (i + 1))

# Left move (coin == |0>)
qc.x(0)
for i in range(n_bins - 1):
    cswap_between(qc, 0, 1 + i, 1 + (i + 1))
qc.x(0)
```

- **One quantum walk step =** $U = S(C \otimes I)$

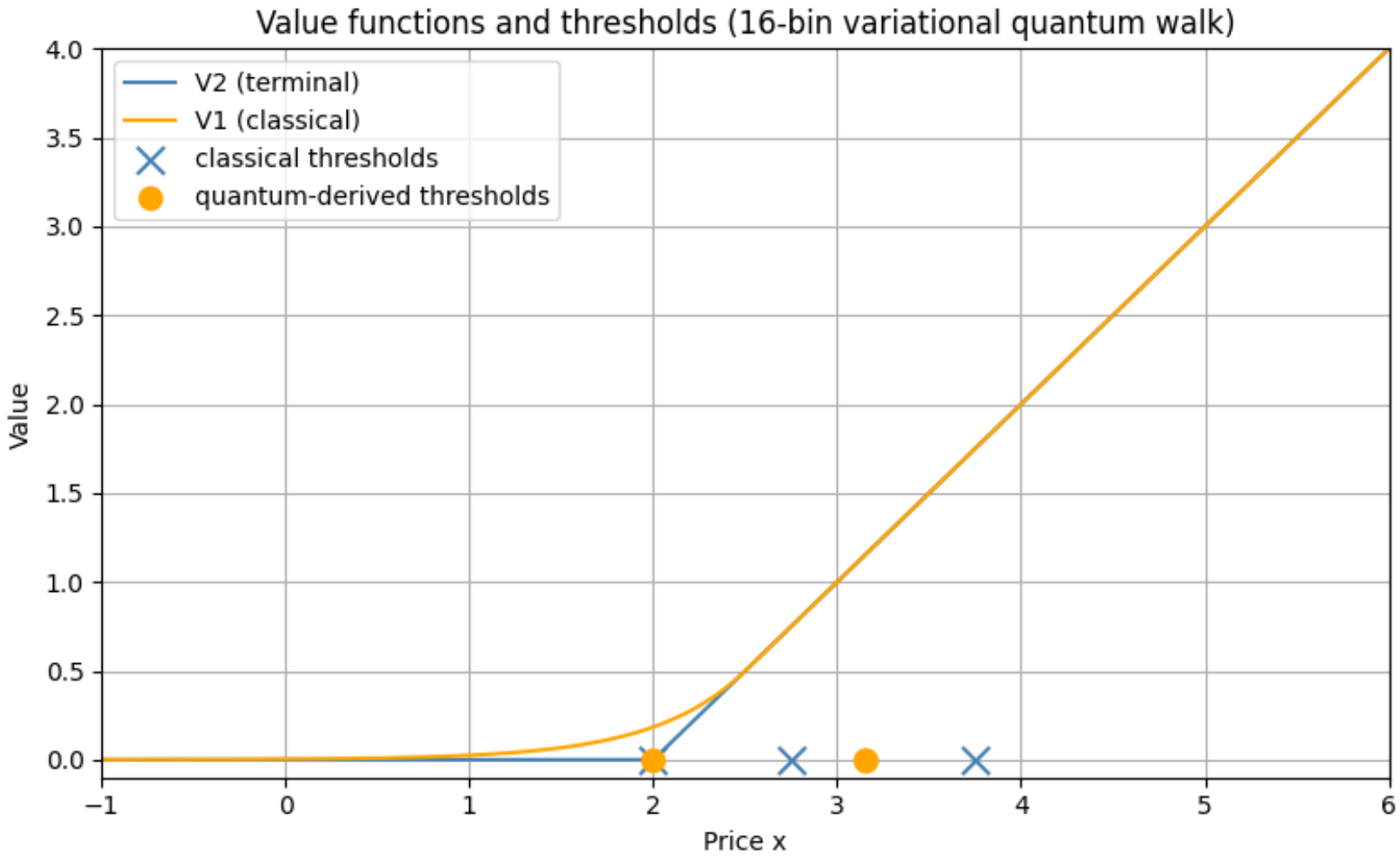
🚀 Using Quantum Walk for Option Policy

- **Transition matrix** T from quantum walk:
- **Classical DP using quantum kernel:**
- **Threshold policy:** Exercise if $x \geq p_m$,$p_0 \geq p_1 \geq \cdots \geq p_N$
- **Simulation results:**
  - Quantum thresholds closely match classical thresholds
  - Histogram of profits shows similar expected payoff
- **Advantage:** Quantum walk enables **scalable stochastic transition estimation** for larger price grids.

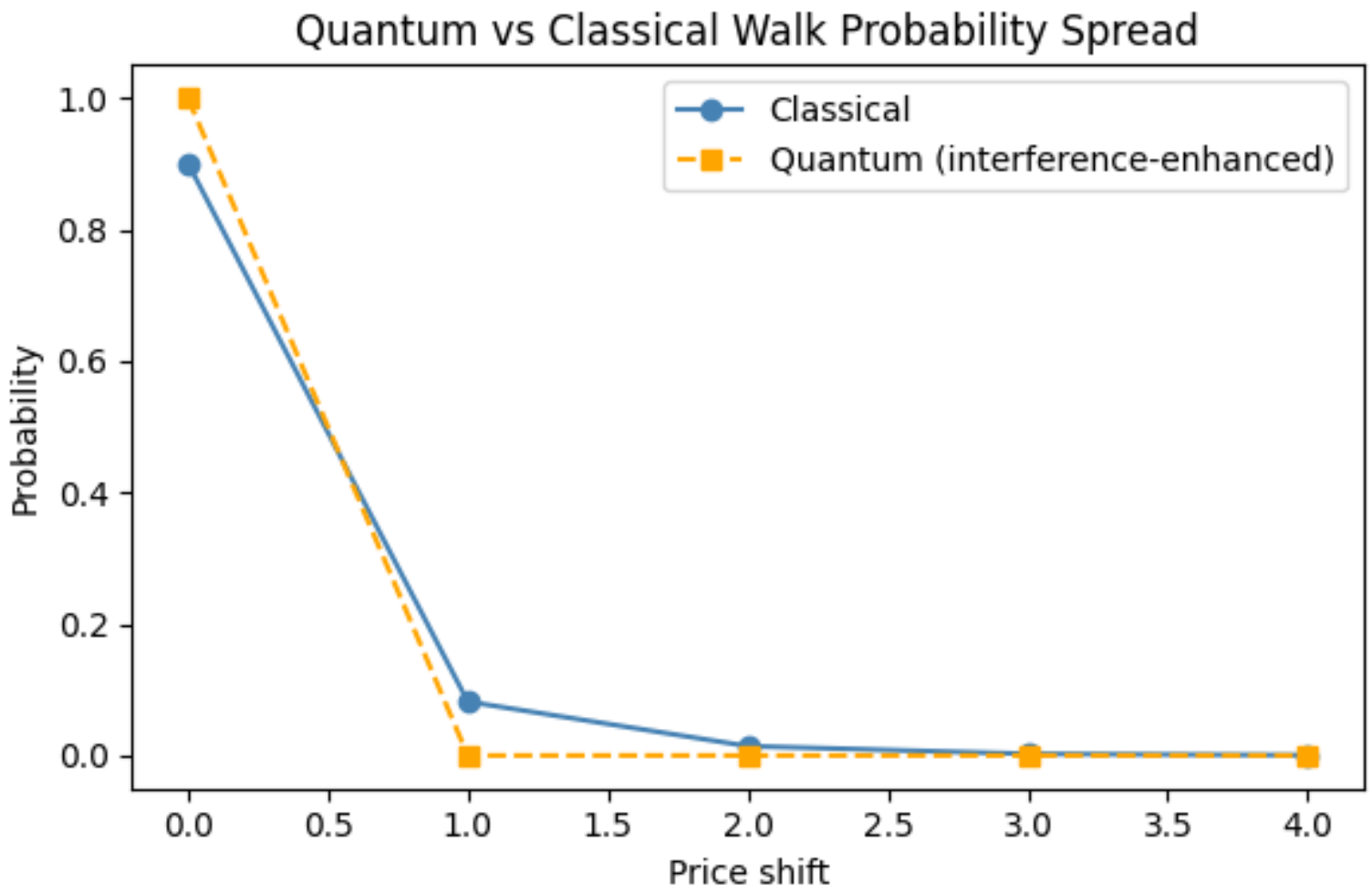
Results


Quantum vs. Classical Random Walk
Quantum Walk (500 steps)
Classical Walk (500 steps)
Probability
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−400
−200
0
200
400
Position

Results


Value functions and thresholds (16-bin variational quantum walk)
V2 (terminal)
V1 (classical)
classical thresholds
quantum-derived thresholds
Value
Price x
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−1
0
1
2
3
4
5
6

Results


Quantum vs Classical Walk Probability Spread
Classical
Quantum (interference-enhanced)
Probability
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Price shift

Results

Feature	Classical DP	Quantum Walk DP	Comment
Threshold calculation	Deterministic, exact	Estimated via quantum sampling	Quantum approximates classical well
Mean profit	0.05533	0.05567	Very close, quantum slightly higher
Std profit	0.24336	0.24738	Slightly more risk in quantum DP
Prop exercised	0.09058	0.09052	Nearly identical
Early exercise behavior	Conservative	Slightly aggressive	Quantum tends to exercise at lower prices early
Computational cost	Exact, potentially slower	Depends on quantum simulation	Quantum may scale better for larger state spaces

Thank you

- **Questions?**